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MAGNETOSTRATIGRAPHY AND CORRELATION OF MIOCENE BASALTS OF THE NORTHERN OREGON COAST AND COLUMBIA PLATEAU, SOUTHEAST WASHINGTON

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ABSTRACT. Paleomagnetic measurements are used to supplement other techniques in an attempt to resolve stratigraphic problems in the Columbia River Basalt Group of the Columbia Plateau in southeast Washington and in basalts exposed along the northern Oregon coast. During the middle and late Miocene, the coastal and plateau basalts were erupted from separate sources hundreds of kilometers apart. We have collected 384 oriented cores from 63 sites for purposes of developing the magnetostratigraphy and determining the temporal relationships of the coastal basalts with those exposed on the Columbia Plateau. This survey studied dikes and flows from members of the Saddle Mountains, Wanapum, and Grande Ronde Basalts of the Yakima Basalt Subgroup on the plateau and from the Cape Foulweather and Depoe Bay Basalts of the Oregon coast.

Widely divergent virtual geomagnetic pole (VGP) positions characterize different members of the Saddle Mountains Basalt, the youngest formation in the Columbia River Basalt Group. These divergent VGPs are distinctive stratigraphic markers and should prove useful for future magnetocorrelation of these flows.

Although previous results showed the Roza Member (Wanapum Basalt) to be of transitional magnetic polarity (Rietman, ms), new evidence suggests that the southern part of its vent system was active during the earliest part of the subsequent reversed polarity epoch.

A detailed magnetostratigraphy of Grande Ronde Basalt exposed in the Blue Mountains of southeast Washington shows that the 400-m thick section was extruded during three successive polarity epochs and can be divided into three corresponding magnetostratigraphic units (magnetozones), from oldest to youngest, N1, R2, and N2. An older reversed unit, R1, is present in nearby areas.

Unusual VGP positions characterize both the Cape Foulweather Basalt of coastal Oregon and a few flows of the chemically similar Frenchman Springs Member of the Wanapum Basalt (Rietman, ms). The flows may have been extruded during the same geomagnetic excursion.

Two unusual paleopole positions characterize the Depoe Bay Basalt of the Oregon coast and permit its tentative correlation with two successive Grande Ronde flows in southeast Washington. This refines the previously vague temporal relationship of the two units.

The slight westerly paleodeclination of the Yakima Basalt precludes major rotational deformation in southeast Washington since the middle Miocene.

INTRODUCTION

Large volumes of tholeiitic lava "flooded" wide areas in northern Oregon, eastern Washington, and western Idaho between about 16.5 and 6 m.y. before present (m.y.b.p.), creating the Columbia Plateau (fig. 1). The basalt flows, known collectively as the Columbia River Basalt Group, are exposed over an area exceeding 2×10^5 km^2 and have a minimum estimated volume of 2×10^5 km^3 (Waters, 1962). Recent investigations have resulted in a regionally applicable stratigraphic subdivision of the basalt (fig. 2; Swanson and others, 1979). In this study, paleomagnetism supplements other criteria, including geochemistry, petrography, and field observations, in an effort to characterize important stratigraphic units in the Saddle Mountains, Wanapum, and Grande Ronde Basalt, and to evaluate possible correlations of single flows across wide areas.

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Sampling was conducted in southeast Washington, an area traversed by the Snake River, whose canyon walls afford good access to the basalt (fig. 3). The region also includes part of the Blue Mountains, an uplifted area in which more than 800 m of Grande Ronde Basalt is exposed. A geologic map of the area has recently been completed (Swanson and others, 1977) and served as a guide to the sampling.

We sampled most stratigraphic units at sites tens of kilometers apart to determine if chemically-identical flows are coeval. These include flows that spread as sheet floods across the plateau and those that partly filled ancestral Snake River canyons cut during the waning stages of volcanism, 13 to 6 m.y.b.p.

We also investigated the correlation of a few flows with possible feeder dikes. These data are relevant to ideas regarding the timing of

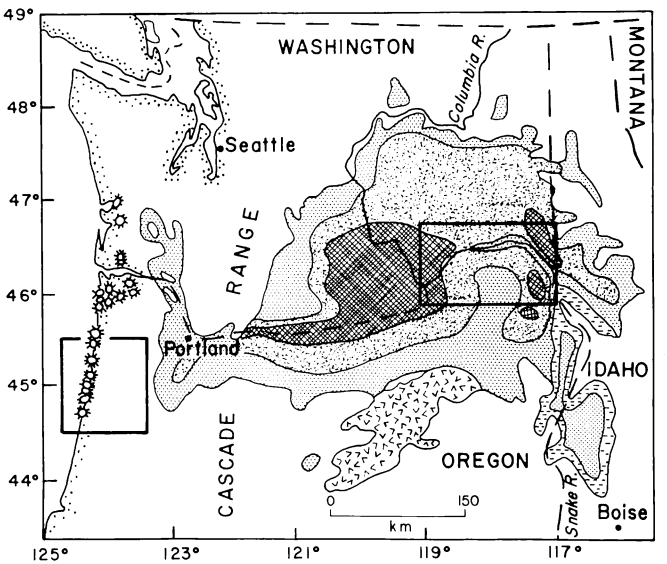


Fig. 1. Generalized distribution map of stratigraphic units in the Columbia River Basalt Group and locations of coastal volcanic centers (modified from Wright, Grolier, and Swanson, 1973; Snavely, MacLeod, and Wagner, 1973; Swanson and others, 1977).

coastal eruptive centers

Saddle Mountains Basalt

Wanapum Basalt

Grande Ronde Basalt

Picture Gorge Basalt

Innaha Basalt

detail—fig. 7

detail—fig. 3

Note: Saddle Mountains Basalt is also exposed in isolated outcrops along the Snake River from southeast Washington to Devils Canyon (see fig. 3).

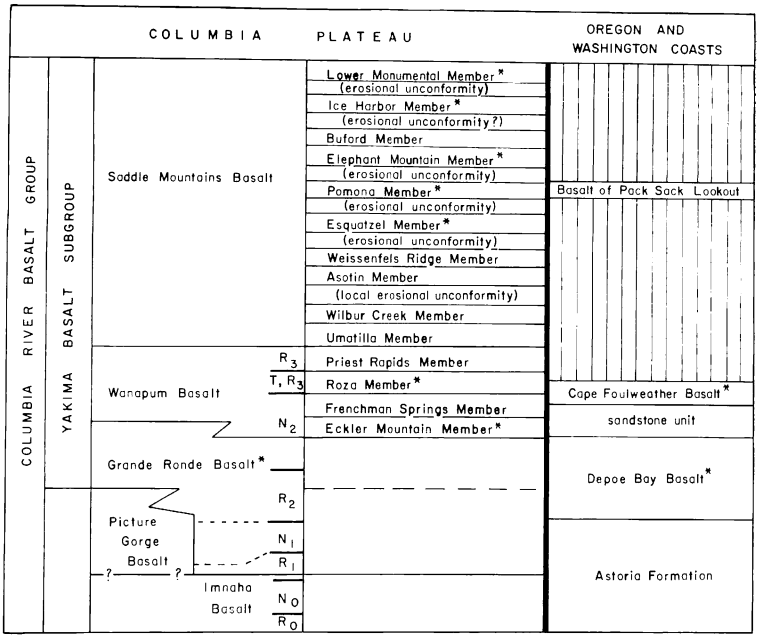


Fig. 2. Stratigraphic relationships in the Columbia River Basalt Group, Oregon, Washington, and Idaho, and in the basalts of the Coast Range, Oregon and Washington. Magnetozone assignments for the Wanapum Basalt and underlying formations are also included: N = normal, T = transitional (θ<45°), R = reversed polarity. Note: the "oldest intracanyon flows," which occur between the Esquatzel and Priest Rapids Members, is not a formally recognized name. * Indicates sampled members (modified from Snavely, MacLeod, and Wagner, 1973; Swanson and others, 1979).

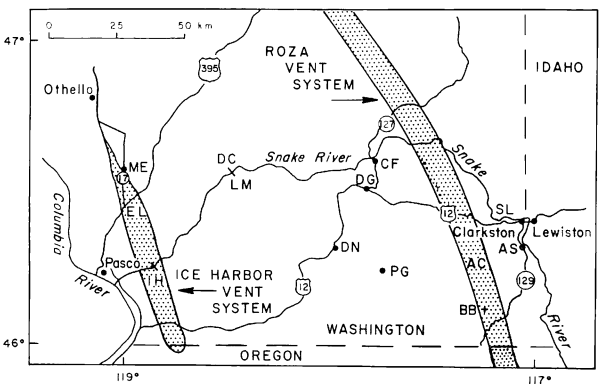


Fig. 3. Sampling sites in the Columbia River basalt and localities mentioned in the text. Key: AC, Asotin Creek; AS, Asotin; BB, Big Butte; CF, Central Ferry; DC, Devils Canyon; DG, Dodge; DN, Dayton; EL, Eltopia; IH, Ice Harbor Dam; LM, Lower Monumental Dam; ME, Mesa; PG, Patrick Grade; SL, Silcott.

activity along, and geometry of, segments of vent systems for single flows.

Local volcanic centers in coastal Oregon and possibly coastal Washington (fig. 1) produced small volumes of basalt geochemically and petrographically identical to, and in the same stratigraphic order as, two and possibly three of the principal stratigraphic units on the Columbia Plateau (fig. 2; Snavely, MacLeod, and Wagner, 1973). Based on K–Ar and fossil evidence (Turner, 1970; Holmgren, 1970; Gray and Kittleman, 1967; Evernden and James, 1964), the coastal and plateau basalts appear coeval, but resolution of ages by these techniques is not good. We therefore sampled the Cape Foulweather and Depoe Bay Basalts to investigate possible time-equivalency with their apparent correlatives on the Columbia Plateau, the Frenchman Springs Member of the Wanapum Basalt, and the Grande Ronde Basalt, respectively.

Finally, we interpreted our data with regard to the orocline hypothesis (Carey, 1958), which attributes many structures of the Pacific Northwest to post-orogenic deformation.

This paper builds on the work of several others, particularly Rietman (ms), Watkins (1964, 1965), and Watkins and Baksi (1974). A concurrent study by Hooper and others (1979) deals with the magnetostratigraphy of the Grande Ronde and Imnaha Basalt in western Idaho.

This paper is dedicated to the memory of Norman Watkins, whose pioneering, albeit controversial, paleomagnetic work on the basalt of the Pacific Northwest has provided a stimulus for all subsequent investigations.

PRINCIPLES OF MAGNETOCORRELATION OF LAVA FLOWS

Some initial assumptions required in magnetocorrelation studies are: (1) on cooling, the entire volume of a flow acquires a magnetization parallel to the ambient geomagnetic field; this primary remanent direction can be isolated by appropriate laboratory techniques; (2) the cooling rate of a flow unit is fast relative to the rate of paleosecular variation (PSV) of the geomagnetic field, thus enabling its entire volume to record the same field direction; and (3) the frequency of extrusion of individual cooling units was low relative to the rate of PSV, thus giving each flow a high probability of acquiring a unique magnetic signature within a polarity epoch.

The third assumption may not be strictly correct, since the geomagnetic field may have had similar orientations at various times within a single polarity epoch. Unless PSV is of low amplitude, however, the probability of flows of different ages acquiring the same magnetic signature is typically less than 1 percent (Cox, 1971; Grommé, McKee, and Blake, 1972). No definitive study of Miocene secular variation in the Pacific Northwest has been reported, but our data from the Grande Ronde Basalt (table 2) suggest that PSV was of sufficient magnitude to justify assumption 3.

If two flows of similar chemical composition, mineralogy, and stratigraphic position feature distinctly different VGPs (that is, no overlap of confidence ovals; Fisher, 1953), they cannot be products of the same eruption. Because of the random nature of secular variation, angular deviations between dissimilar VGPs cannot be used to estimate age differences of the flows. Thus, two flows with different VGPs but with otherwise similar characteristics may differ in age by a few years or millions of years.

If other correlation criteria are satisfied, overlapping confidence ovals of two high latitude (>45°) VGPs suggest that the flows were erupted over a short period of time relative to geomagnetic secular variation; that is, within a few centuries or less (Grommé, McKee, and Blake, 1972). Similarly, identical low latitude (<45°) VGPs imply contemporaneity with perhaps greater probability, since the geomagnetic field has such orientations infrequently and for extremely brief intervals (Cox, Hillhouse, and Fuller, 1975). Polarity transitions occur rapidly — probably in less than 5000 yrs (Ninkovich and others, 1966; Opdyke, 1972; Rikitake, 1972). Polar movement, therefore, is greatly accelerated in a transitional geomagnetic regime.

An excursion refers to migration of the VGP to the opposite hemisphere for a period of perhaps only 10^2 to 10^3 yrs (Geotimes, 1976, April, v. 21, p. 18-20). Because of the limited resolution of the paleomagnetic method (Cox, 1975) and the incomplete nature of the resulting record, the frequency of geomagnetic excursions will probably never be known. Amerigian (in preparation), however, recognizes 82 distinct excursions over a 9 m.y. period in Cenozoic lava flows in Iceland.

METHODS

Field.—In July 1975, we collected 384 cores from 63 sites: 12 in basalt along the northern Oregon coast and 51 in the Columbia River basalt and associated feeder dikes in southeast Washington (fig. 1). An average of 6 cores was removed from each outcrop, using a portable gasoline-powered drill (Doell and Cox, 1964). About half the coastal sites were oriented by siting to features identifiable on quadrangle maps; all other sites were oriented with a sun compass. Samples were collected from artificial exposures as well as natural outcrops (see app). In most flows, individual samples were drilled at horizontal intervals of a few meters. Coring was more closely spaced in narrow dikes.

Laboratory.—One sample (2.5 × 2.2 cm) was sliced from the deepest, least weathered part of each core. The direction and intensity of natural remanent magnetism were measured on a Digico spinner magnetometer. All specimens were progressively demagnetized in peak alternating fields of at least 200 oe. In nearly all cases, the stable primary component was isolated at, or before, the 300 oe treatment, but several samples required peak alternating fields as high as 600 oe to minimize secondary components. Specimens from a few sites were also subjected to thermal demagnetization at temperatures up to 450°C in a Schonstedt Thermal Specimen Demagnetizer.

Paleomagnetic data were reduced using Fisher (1953) statistical methods. A minimum scatter criterion (Watkins and Richardson, 1968) was utilized to define the primary remanent vector for each site. Since the coercivity spectra for all samples from a single site are rarely the same (Watkins and Haggerty, 1967; Wilson, Haggerty, and Watkins, 1968), it is reasonable to assume that an alternating field of a given peak value may not sufficiently "clean" all specimens from a given site. Therefore, in this study, the site mean direction is the summation of those vectors (one per sample, regardless of demagnetizing treatment) which produce a maximum resultant vector.

The final mean direction of remanent magnetism of each coastal flow was corrected for its observed dip (N. S. MacLeod, written commun., 1976), assuming that the flows were originally horizontal. The assumption seems valid, as the slope of the pre-basalt terrain (coastal plain-continental shelf) was probably less than two or three degrees. The cor-

TABLE 1

A. Saddle Mountain Basalt

Member (flow)	Site	N	D	I	α_{95}	θ	ϕ	Polarity
Lower Monumental	LM 1	5	341.8	+71.2	3.5	+75.2	198.3	N
	LM 2	4	334.6	+55.0	9.8	+68.1	131.2	N
Ice Harbor (Goose Island)	GI 1	6	51.3	-19.2	11.7	+17.5	8.3	T
	GI 2	6	33.5	-31.9	13.3	+19.5	28.0	T
(Martindale)	MA 1	8	185.1	-55.1	3.0	-78.6	220.5	R
	MA 2	8	168.2	-65.0	3.3	-81.9	340.2	R
(Basin City)	BC 1	5	317.0	+54.7	4.2	+56.1	148.4	N
	BC 2	6	319.3	+49.5	6.3	+54.8	139.4	N
	BC 3	6	24.0	+68.5	1.7	+73.5	304.0	N
Elephant Mountain	EM 1	6	6.9	+59.6	4.2	+82.2	19.8	N
	EM 2	4	14.2	+62.5	8.3	+79.7	340.9	N
	EM 3	6	115.2	-25.6	9.3	-27.0	323.0	T
	EM 4	6	127.9	-39.5	9.8	-41.9	320.5	T
Pomona	PO 1	6	187.8	-51.9	1.7	-74.9	215.9	R
	PO 2	6	186.3	-53.4	2.2	-76.6	218.6	R
	PO 3	6	203.4	-54.6	2.0	-69.1	176.4	R
	PO 4	6	195.4	-50.6	2.9	-70.8	198.3	R
Esquatzel	EQ 1	6	340.8	+63.0	3.6	+76.4	150.6	N
	EQ 2	6	308.0	+81.7	11.4	+54.5	219.5	N
Oldest i. f.	OI 1	5	196.3	-45.5	6.0	-66.7	202.8	R

B. Wanapum Basalt

Member (flow)	Site	N	D	I	α_{95}	θ	ϕ	Polarity
Roza	RZ 1	5	214.2	-14.7	7.3	-41.3	194.2	T
	RZ 2	6	184.7	-38.7	7.0	-65.2	231.6	R
Eckler Mountain (Dodge)	DJ 1	6	3.5	+62.6	1.2	+86.6	14.9	N
	DJ 2	6	0.9	+59.5	3.6	+83.9	55.6	N
(Robinette Mountain)	RM 1	5	344.2	+62.5	6.5	+78.6	145.0	N

N = number of separately oriented cores. D and I = declination and inclination of remanent magnetism (after demagnetization treatment) in degrees east of north and in degrees with respect to the horizontal (+ = down, − = up). α_{95} = semivertical angle of 95 percent confidence cone. θ and ϕ = latitude and longitude of virtual geomagnetic pole (VGP). θ is (+) for northern, and (−) for southern hemisphere. ϕ is the degrees east of Greenwich. Polarity is normal (N), transitional (T = VGP lat $< 45°$), or reversed (R).

TABLE 2

Grande Ronde Basalt
Patrick Grade

Flow no.	N	D	I	α_{95}	θ	ϕ	Polarity
2	7	342.8	+49.2	1.8	+68.9	107.3	N
3	6	359.5	+53.2	4.8	+77.4	63.9	N
4	6	358.6	+64.6	2.3	+89.0	157.5	N
5	6	354.1	+63.7	3.4	+85.7	139.7	N
6	6	4.2	+56.5	7.3	+80.2	41.8	N
7	6	340.7	+66.1	3.3	+76.8	167.8	N
8	6	331.5	+21.6	4.6	+47.3	105.7	N
9	6	261.4	-86.2	7.8	-47.0	73.0	R
10	6	339.0	-50.2	19.1	+10.3	80.2	T
11	5	355.9	+63.1	12.0	+86.6	121.2	N
12	6	183.5	-69.0	4.4	-83.5	81.3	R
13	6	201.7	-59.9	3.7	-73.4	163.7	R
14	6	206.7	-74.3	5.5	-68.9	99.8	R
15	6	203.0	-71.8	1.6	-72.6	107.6	R
16	6	185.9	-64.8	4.3	-85.9	144.7	R
17	6	192.6	-64.6	2.8	-81.3	146.7	R
18	6	200.2	-74.5	6.8	-71.3	93.3	R
19	6	183.7	-72.0	2.1	-79.2	73.0	R
20	5	87.9	-78.7	4.9	-41.5	32.2	T
21	6	65.0	+ 6.4	9.8	+19.4	348.5	T
22	6	317.9	+69.8	2.2	+62.4	182.9	N
23	6	328.6	+72.5	3.2	+67.9	194.4	N
24	6	328.5	+63.1	7.3	+68.0	158.8	N

For explanation of symbols, see caption to table 1. Flow numbers are in stratigraphic order; increasing number = increasing age.

TABLE 3

Coastal Basalts

Stratigraphic unit	Site	N	D	I	α_{95}	θ	ϕ	Polarity
Cape Foulweather Basalt	CF 1	6	159.9	+36.2	4.3	-22.3	256.4	T
	CF 2	6	198.7	+53.5	9.7	- 9.1	220.3	T
Depoe Bay Basalt	DB 3	6	356.8	+22.7	12.2	+56.7	61.7	N
	DB 4	6	344.0	+21.1	5.8	+53.2	82.9	N
	DB 5	6	345.4	+13.9	5.6	+45.9	78.8	N
	DB 6	5	336.6	+30.5	11.2	+55.3	98.0	N
	DB 8	6	354.4	+22.5	6.4	+56.3	65.9	N
	DB 1	4	274.6	-80.4	10.0	-40.6	80.9	T
	DB 2	5	245.0	-80.7	8.6	-49.9	[illegible]	R

For explanation of symbols, see caption to table 1.

rections were based on three-point solutions to the basal contact of the basalt units and are accurate within a few degrees. Results from a single plateau site (PO 4, Pomona Member) were similarly corrected; the remaining sites did not require tectonic correction. Tilt corrections are listed in the appendix.

RESULTS AND DISCUSSION

The results are presented and discussed in descending stratigraphic order, first for the plateau flows and then for the coastal basalts. Mean paleomagnetic data for each site comprise tables 1 to 3, and site mean pole positions are plotted on north-polar projections in figures 4, 5, and 8.

A. Paleomagnetic Properties

A site is termed magnetically stable if the magnetic vector of most samples acquires a stable orientation at some point in the demagnetization treatment, that is further demagnetization produces intensity and not directional, changes. Many sites feature median destructive fields of less than 100 oe, but a smooth decrease in intensity follows the initial sharp intensity drop.

Reversely-magnetized samples often show an increase in intensity during the initial stages of demagnetization, followed by a smooth decrease on further cleaning. In all cases, it may be shown vectorially that the primary increase is due to the removal of a "soft" normal-polarity overprint acquired during the last 690,000 yrs (the Brunhes geomagnetic epoch).

A few sites (GI 2, PG 5, PG 6, and PG 7) included both normally- and reversely-magnetized samples even after demagnetization in peak alternating fields of 300 oe. Subsequent thermal demagnetization at 400°C was successful in minimizing the high coercivity secondary components in those samples.

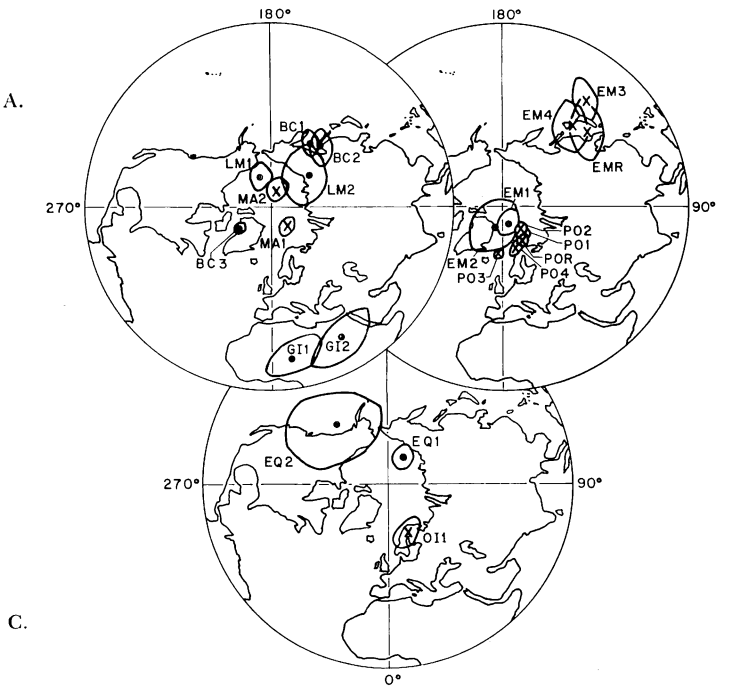


Fig. 4. Virtual geomagnetic poles (VGP) and associated confidence ovals (α_{95}) of the Saddle Mountains Basalt, as compiled in table 1A: • = normal polarity; X = reversed polarity, transposed to the northern hemisphere. (A) Lower Monumental (LM) and Ice Harbor (GI, MA, and BC) Members. (B) Elephant Mountain (EM) and Pomona (PO) Members; EMR and POR, Rietman's (ms) mean poles. (C) Esquatzel Member (EQ) and the oldest intracanyon flows (OI).

A two part rejection criterion was applied to the data: (1) the site must be magnetically stable, as defined above, and (2) the semi-angle of the 95 percent confidence cone (α_{95}; Fisher, 1953) about the site mean VGP must not exceed 25°.

Three magnetically unstable sites were rejected: one each in the Roza Member of the Wanapum Basalt and the oldest intracanyon flows of the Saddle Mountains Basalt and one in the Depoe Bay Basalt of the Oregon coast. Part two of the rejection criterion resulted in deletion of three additional sites: one in the Lower Monumental Member of the Saddle Mountains Basalt and two in the Cape Foulweather Basalt of coastal Oregon.

Individual samples were also omitted from a few sites if the direction of magnetization was deflected through an angle >45° from an otherwise tight cluster of directions. Of the 57 sites retained for analysis, only three have mean directions defined by as few as four samples (tables 1-3).

B. Brief Description and Magnetostratigraphy of Each Member

1. Saddle Mountains Basalt.—Volcanism that produced the Saddle Mountains Basalt was highly intermittent and of much greater duration (from about 13 to 6 m.y.b.p.) than earlier phases of Yakima volcanism. The Saddle Mountains Basalt is chemically and isotopically more heterogeneous than other formations in the Columbia River Basalt Group (Wright, Grolier, and Swanson, 1973; Swanson and others, 1979). The flows probably account for less than 1 percent of the total volume of Columbia River basalt. Sedimentary interbeds and erosional unconformities are common throughout the formation (fig. 2).

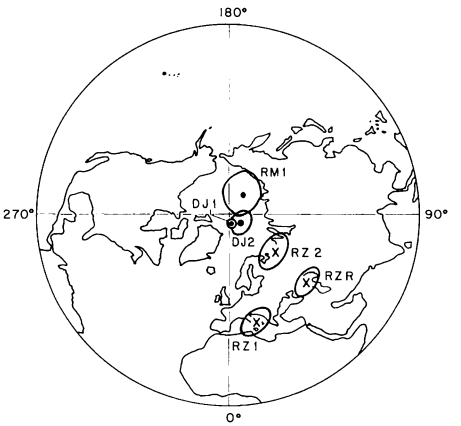


Fig. 5. Virtual geomagnetic poles (VGP) and associated confidence ovals (α_{95}) of the Wanapum Basalt, as compiled in table 1B. Roza (RZ) and Eckler Mountain (DJ, RM) Members. RZR = Rietman's (ms) Roza pole. • = normal polarity; X = reversed polarity, transposed to the northern hemisphere.

1. *Lower Monumental Member*: These 6 m.y.b.p. flows (McKee, Swanson, and Wright, 1977) were confined to the ancestral Snake River canyon and have been mapped along 140 km of its extent (Swanson and others, 1977). The unit averages 20 m in thickness, with a maximum preserved thickness of 60 m.

Sites at Lower Monumental Dam and south of Central Ferry (LM 1 and LM 2, table 1A and fig. 4A) yield different VGPs. This conflicts with field, geochemical, and petrographic data which suggest that the Lower Monumental is a single-flow cooling unit. Local vents have not been found, and the source of the basalt is interpreted to be east of the sampling sites (Swanson and others, 1979). Since the lava moved westerly downcanyon, one might expect that two or more successive flows would somewhere appear in a single outcrop, but this is not the case despite close inspection of the 41 known remnants of the members.

The paleomagnetic data suggest that at least two normally-magnetized flows comprise the Lower Monumental Member. In view of the geologic arguments for only one flow, more sampling is desirable.

2. *Ice Harbor Member*: These flows, confined to the central part of the Columbia Plateau, were erupted from local vents during the waning stages of volcanism about 8.5 m.y.b.p. (McKee, Swanson, and Wright, 1977). They are nowhere more than 30 m thick and generally less than 15 m thick. Their aggregate volume probably does not exceed 10 km^3 (Swanson, Wright, and Helz, 1975).

i. *Basalt of Goose Island* (formerly Ice Harbor 2): The youngest of the four chemical types is characterized by the phenocryst assemblage olivine + magnetite + plagioclase. Dike GI 2 at Eltopia is a differentiate of magma similar to that that produced the Goose Island flow (GI 1) sampled along the Snake River near Ice Harbor dam (Helz, ms). Their low-latitude pole positions (fig. 4A) are indistinguishable and suggest contemporaneity. Thus the flow and dike are probably related to the same eruptive event.

ii. *Basalt of Martindale (formerly Ice Harbor 1)*: These flows bear the phenocryst assemblage olivine + augite + plagioclase and carry large clots of augite + plagioclase. The basalt may be further divided into two subtypes on the basis of phenocryst abundance and small chemical differences (Helz, Wright, and Swanson, 1976; Helz, ms). Flows of both subtypes sampled near Ice Harbor dam (fig. 3) are reversely magnetized and are represented by the distinctly different poles MA 1 (younger) and MA 2 (table 1A and fig. 4A).

iii. *Basalt of Basin City*: The oldest Ice Harbor flows have an olivine + plagioclase phenocryst assemblage (Helz, ms). Most flows fill a narrow graben, generally about 15 m deep and less than 150 m wide, developed along the northern part of the vent system (Swanson, Wright, and Helz, 1975).

The coincident VGPs for flow(s) near Eltopia (BC 1 and BC 2; table 1A and fig. 4A) suggest that the two sites are in either a single cooling unit or two separate units erupted in rapid succession. The

chemistry and petrography of a dike near Ice Harbor dam (BC 3) indicate that it is related to, but slightly different from, flows BC 1 and BC 2 (Helz, ms). Although all three sites have normal polarity, the magnetic data (table 1A and fig. 4A) demonstrate that the dike and flows are not strictly coeval, consistent with the slight chemical and petrographic differences.

These results indicate the occurrence of at least two intrusions of Basin City magma — one that produced flows BC 1 and BC 2, and one that formed dike BC 3.

3. *Elephant Mountain Member*: This unit is essentially aphyric and has an average thickness and minimum volume of 10 m and 103 km^3, respectively (Schmincke, 1967). Several flows in the Elephant Mountain Member are known to occur in the Mesa-Othello area of the central plateau. At least one flow occurs along the ancestral Snake River canyon between Mesa and Asotin (Swanson and others, 1977). The absence of known vents in the central plateau suggests that all Elephant Mountain flows poured westward down ancestral drainages from vents in the eastern plateau. The unit, dated at about 10.5 m.y. (McKee, Swanson, and Wright, 1977), was sampled at four locations. Intracanyon remnants were drilled north of Asotin (EM 1) and at Devils Canyon (EM 2), and two flows on the plateau surface were sampled at Mesa (EM 3) and at Ice Harbor (EM 4; fig. 3).

The two intracanyon remnants have similar magnetic signatures, indicating that the same flow may have been sampled at both sites, located more than 140 km apart along the Snake River. These normally-polarized flows have a distinctly different magnetic signature than the plateau flows, which have transitional polarity (table 1A and fig. 4B). Rietman's (ms) results from four sites in the Elephant Mountain Member on the plateau surface yield a mean VGP (lat = −37.6°, long = 312.2°, α_{95} = 12.1°) which is identical to our data from the two plateau sites (fig. 4B).

The data suggest that the intracanyon flow is not a precise time correlative with any of the flows on the plateau surface sampled by Rietman and us. Not all the Elephant Mountain flows on the plateau surface have been subjected to paleomagnetic analysis, however. Thus, it is possible that one or more of those flows is the precise time equivalent of the intracanyon flow(s). Should further sampling continue to show differences, vents for flows on the plateau surface may be hidden in the central part of the plateau.

4. *Pomona Member*: This unit, the most widespread and easily recognized member in the Saddle Mountains Basalt, was defined and described by Schmincke (1967). The member, dated at about 12.0 m.y. (McKee, Swanson, and Wright, 1977), has an average thickness and minimum volume of 30 m and 600 km^3, respectively (data modified from Schmincke, 1967). The Pomona has been considered, on geologic, geochemical, and isotopic grounds, to be a single flow erupted at an un-

known site in the extreme eastern plateau, probably in western Idaho (Swanson and others, 1979).

This extensive member was sampled at four localities (fig. 3): on the plateau at Mesa (PO 1), in the ancestral Snake River canyon at Devils Canyon (PO 2), near Central Ferry (PO 3), and at Silcott (PO 4). The unit has reversed polarity, and the good cluster of pole positions suggests that the Pomona was erupted over a single short time interval. The small scatter of VGPs may be the result of secular variation during migration of the Curie point isotherm away from the flow margin as the thick intracanyon flow (>100 m in places) cooled over a period of perhaps 100 yrs. Rietman's (ms) data agree excellently with ours (fig. 4B).

Our work correlates the Pomona Member across approx 175 km of southeast Washington and together with Rietman's (ms) data helps correlate a single flow between Silcott and Mosier, Oregon, a distance of about 350 km.

5. *Esquatzel Member*: This unit is characterized by both plagioclase and clinopyroxene phenocrysts and glomerophyric clots less than 5 mm in diameter. Flows averaging 10 m in thickness (max 50 m) are confined mostly to paleocanyons from Mesa more than 70 km east up the ancestral Snake River canyon (fig. 3). We sampled the Esquatzel at two localities approx 40 km apart (fig. 3): at Mesa, near the mouth of the old canyon (EQ 1), and at Devils Canyon, within the old canyon (EQ 2). The paleomagnetic data (table 1A and fig. 4C) are evidence that the Esquatzel may consist of two flows. The Devils Canyon site, however, is in a pillow basalt and hyaloclastite complex, which are unstable bodies that commonly slump during emplacement or cooling. If that happened after some parts of the flow had acquired a thermoremanent magnetization, within-site and between-site directional scatter would result. Nearby sites of unquestioned stability could be sampled in order to verify this hypothesis.

6. *Oldest intracanyon flows*: These flows are plagioclase-phyric with an irregular distribution of 1-cm long crystals. Small differences in TiO_2 and other oxides suggest that the unit comprises at least two flows. They occur in a few isolated remnants in the ancestral Snake River canyon from Devils Canyon upriver to Asotin (fig. 3).

We sampled two of those remnants. Data from one site are unsuitable for paleomagnetic analysis (see "Paleomagnetic Properties"). The remaining site, at Devils Canyon (OI 1), has reversed polarity (table 1A and fig. 4C).

II. Wanapum Basalt.—This formation, which accounts for approximately 5 percent of the total volume of the Yakima Basalt Subgroup, is chemically more homogeneous than the younger Saddle Mountains Basalt. The Wanapum is commonly separated from the underlying Grande Ronde Basalt by either a prominent sedimentary interbed (Mackin, 1961; Bond, 1963) or a saprolite horizon. Most flows of Wanapum Basalt belong to one of three major chemical types, which together cover the same MgO range as the underlying Grande Ronde

flows but which have considerably lower SiO_2 and higher iron and TiO_2 relative to MgO (Wright, Grolier, and Swanson, 1973). The Eckler Mountain Member contains flows of three different compositions, all of which bear some resemblance to compositions of the older Grande Ronde Basalt. No important periods of erosion have been recognized within the formation, and sedimentary interbeds are rare.

Paleomagnetic evidence, presented below, also suggests relatively rapid eruption of the Wanapum Basalt. Data for the Roza Member suggest that it records the transition to the polarity epoch recorded by the overlying Priest Rapids Member. Thus, the Wanapum Basalt probably includes successive magnetostratigraphic units (magnetozones), N2 and R3 (fig. 2).

1. *Roza Member*: The Roza Member is one of the most widespread stratigraphic units in the Columbia River Basalt Group. Swanson, Wright, and Helz (1975) estimate an original extent and volume of 40,000 km^2 and 1500 km^3, respectively. The flows were erupted about 15 m.y.b.p., based on unpublished K–Ar ages and comparison with the dates of Watkins and Baksi (1974). The Roza is characterized by abundant plagioclase phenocrysts which are of single-crystal habit and typically greater than 5 mm in length. The member usually consists of one or two thick flows, with thin flows confined to near-vent areas. A variety of evidence suggests that the Roza was erupted in a short period of time, perhaps of the order of a few hundred to a few thousand years for the entire member: interflow sediments are absent, evidence of significant erosion between successive flows is lacking, and delicate flow-surface textures are commonly preserved (Swanson, Wright, and Helz, 1975). A well-defined fissure system for the Roza Member has been mapped in southeast Washington (fig. 3; Swanson, Wright, and Helz, 1975). The zone, which trends north-northwest and parallels other dikes in the Chief Joseph dike swarm, is probably less than 3 km wide and exceeds 150 km in length. Feeder dikes are exposed in many areas along the central and southern parts of the Roza fissure system.

Three sites were drilled in the vent system: two in flows and one in a dike. One of the flows is magnetically unstable and has been omitted from the study (see "Paleomagnetic Properties"). The remaining sites, both along the north fork of Asotin Creek in the southern part of the vent system (fig. 3), supplement those that Rietman (ms) sampled at widespread localities throughout the plateau.

Site RZ 2, a dike, has reversed polarity, and RZ 1, a flow, was apparently extruded during a polarity transition (table 1B, fig. 5). The reversed polarity of the dike was unexpected, as Rietman's (ms) results clearly showed a transitional polarity for widespread Roza flows (fig. 5). We suggest that the Roza transition terminated in the reversed polarity epoch (R3) recorded in the overlying Priest Rapids Member (fig. 2).

2. *Eckler Mountain Member*: i. *Basalt of Dodge*. These coarse-grained plagioclase-phyric basalts were probably erupted from at least three fissure systems. Generally less than 15 m thick, Dodge flows, known

chiefly in the Blue Mountains of southeast Washington and neighboring Oregon, also occur near St. Maries, Idaho (70 km southeast of Spokane; Griggs, 1976) and along the lower Snake River.

We sampled the unit in two areas (fig. 3), near Dodge (DJ 1), and along the Patrick Grade road in the Blue Mountains (DJ 2), where the flow immediately overlies the Grande Ronde Basalt. The sampled flows have normal polarity (table 1B) and yield VGPs which are indistinguishable at the 95 percent confidence level (fig. 5). Thus, we have sampled either a single cooling unit at both localities, approx 55 km apart, or separate flows extruded in rapid succession. The basalt of Dodge is assigned to magnetozone N2 (figs. 2 and 6; Swanson and Wright, 1976).

ii. *Basalt of Robinette Mountain.* This coarse-grained, diktytaxitic basalt immediately underlies the Dodge, but the two units occur in nearly mutually exclusive areas. The Robinette Mountain, the lowermost Wanapum flow, was erupted from a vent system south of Dayton (fig. 3) and is generally less than 15 to 20 m thick.

We sampled the flow at a single site just south of Dayton (RM 1). The Robinette Mountain flow has normal polarity (table 1B and fig. 5) and is assigned to magnetozone N2 (fig. 2; Swanson and Wright, 1976).

III. Grand Ronde Basalt.—The Grande Ronde Basalt is by far the most voluminous formation in the Columbia River Basalt Group. Its composite stratigraphic thickness is at least 1300 m, and it constitutes between 90 and 95 percent of the total volume of the Yakima Basalt Subgroup. The Grande Ronde is characterized by a general lack of phenocrysts, relatively low FeO content, and higher SiO_2 relative to MgO than most other flows of the Columbia River basalt (Wright, Grolier, and Swanson, 1973). Feeder dikes for the Grande Ronde Basalt are widespread: they have been found as far west as the Grand Coulee in the north-central Columbia Plateau (Swanson, Wright, and Helz, 1975) and as far east as Rocky Canyon in west-central Idaho (Holden and Hooper, 1976).

Magnetostratigraphy: Our study has produced a detailed magnetostratigraphy of a 400-m thick section of Grande Ronde Basalt exposed in the Blue Mountains of southeast Washington. Twenty-three flows in a stack of 25 were sampled at exposures along the Patrick Grade (PG) road (fig. 3). The Dodge flow of the Wanapum Basalt (DJ 2) caps the section, which comprises perhaps the upper two-thirds of the Grande Ronde Basalt. The section was selected because previous field measurements with a fluxgate magnetometer had indicated the presence of three magnetostratigraphic units.

Three magnetozones and four reversals are found in the section (fig. 6). The magnetozones are designated, from oldest to youngest, N1, R2, and N2, in accordance with the numbering system of Swanson and Wright (1976). One transitional- and three normal-polarity flows comprise magnetozone N1. The overlying R2 includes a transitional flow at the base overlain by eight reversely-magnetized flows. Flow 11, the base of N2, is overlain by three flows (PG 8, 9, and 10) which were extruded

during an excursion of the geomagnetic field. The top six normal polarity flows of N2 complete the magnetostratigraphy of the section (fig. 6). The excursion in N2 is a valuable magnetostratigraphic horizon. Indeed, as we will discuss in a later section, the proposed Grande Ronde correlative in coastal Oregon, the Depoe Bay Basalt, may have recorded the same field behavior.

The lower part (N1 and R2) of the Patrick Grade section is stratigraphically correlative with Watkins and Baksi's (1974) Snake River Bend section, but because of the nature of their study (polarity only), it is not possible to make detailed comparisons of directional data from the two areas. It is well to note that a major unconformity occurs near the top of their section, and that the upper three or four flows, totaling 60 m in thickness, belong to the Wanapum Basalt and should not be assigned to their magnetozone R1 (equivalent to our R2). Also, Watkins and Baksi's (1974) Grande Ronde section is repeated by faulting, and "Grande Ronde R" magnetozone is actually about half their interpreted thickness.

A final point of interest concerns the excellent agreement between field and laboratory determinations of polarity. Field analyses in the Patrick Grade section produced results similar to those of this paper, except that polarities for PG 8, 9, and 10 were considered as normal. Nearly perfect agreement between laboratory and field determinations

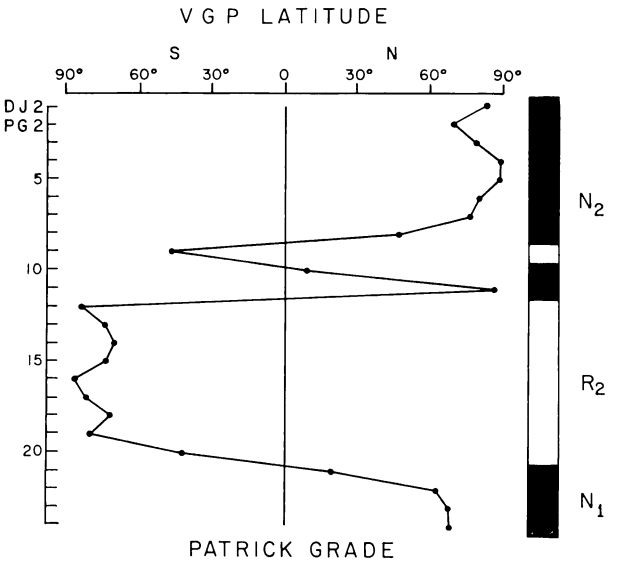


Fig. 6. Magnetostratigraphy of the Patrick Grade section, southeast Washington. All except the top flow (basalt of Dodge) are in the Grande Ronde Basalt. At left, latitude of VGP for each flow, in stratigraphic sequence (stratigraphic number at left, corresponding to table 2). Polarity log at right: black = normal, clear = reversed. Magnetozone assignment is right of log. Single flows lying between flows 11 and 12 and 23 and 24 were not sampled.

exists for all other units of this study. Such good comparison shows that in-situ polarity determinations, if done with care on the more stably magnetized oxidized parts of flows, can be used to establish broad magnetostratigraphic units useful for reconnaissance mapping. The field determinations must, however, be supplemented with laboratory analyses of oriented specimens from critical sections to refine the data.

IV. Basalts of the Oregon Coast.—1. *Cape Foulweather Basalt.* At Cape Foulweather, the upper contact of the Cape Foulweather Basalt is not exposed, and therefore the formation is thicker than the 150 m which are exposed. The large number of intrusive bodies near Cape Foulweather, complex interlayering of extrusive units, and numerous local unconformities indicate that eruptions occurred at several vents (Snavely, MacLeod, and Wagner, 1973). The Cape Foulweather Basalt is of Frenchman Springs chemical type (Wright, Grolier, and Swanson, 1973).

We sampled the formation at 4 sites in 3 localities (fig. 7): at the type locality (CF 1) in a volcanic neck with well-developed radiating columns, at Rocky Point (CF 2) where it is interbedded with pillow breccia and hyaloclastite, and at Yaquina Head. Data from Yaquina Head were rejected (see "Paleomagnetic Properties"). The low latitude positions of 6 VGPs (including 4 additional sites by R. Simpson, ms) in the Cape Foulweather Basalt (fig. 8) suggest that the formation was extruded during a geomagnetic polarity transition or field excursion. If these sites accurately represent the paleomagnetic properties of the entire section, then the Cape Foulweather Basalt must have been erupted at a rapid rate to have recorded a short-lived geomagnetic phenomenon throughout its volume. Such an unusual field orientation is stratigraphically distinctive and has high potential value in correlating Cape Foulweather flows with those of suspected equivalent age east of the Cascade Range.

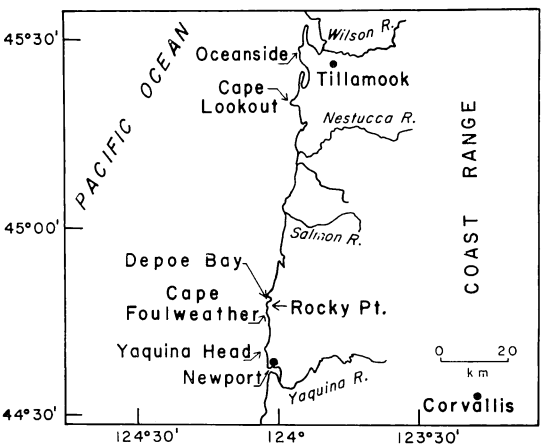


Fig. 7. Sampling localities in the coastal basalt in Oregon: Cape Foulweather Basalt was sampled at Yaquina Head, Rocky Point, and Cape Foulweather; Depoe Bay Basalt was sampled at Depoe Bay, Cape Lookout, and Oceanside.

2. *Depoe Bay Basalt*: The Depoe Bay Basalt is the oldest of the Miocene basalt units in coastal Oregon and Washington. It consists of submarine pillow basalt and breccia, subaerial basalt flows, and dikes, sills, and irregular intrusive bodies. A near-vertical cliff on the south side of Cape Lookout exposes more than 200 m of pillow basalt and columnar-jointed flows underlain and overlain by sediments. The Cape Foulweather Basalt caps the section (Snavely, MacLeod, and Wagner, 1973). K–Ar dates for the Depoe Bay Basalt range from 14 ± 2.7 m.y. (Turner, 1970) to 16 ± 0.65 m.y. (by J. Obradovich, reported *in* Snavely, MacLeod, and Wagner, 1973). The Depoe Bay is chemically similar to the Grande Ronde Basalt on the Columbia Plateau (Snavely, MacLeod, and Wagner, 1973).

We sampled the Depoe Bay Basalt in eight sites at three localities in coastal Oregon (fig. 7). The northernmost sites, at Maxwell Point (DB 1 and DB 2) just north of Oceanside, are exposed at sealevel and are approx 50 m apart. At Cape Lookout, we sampled from five sites in a thick section of flows exposed in roadcuts along the Three Capes Scenic Highway. The stratigraphic positions of DB 3 and DB 7 are uncertain, but observed contact relations show that DB 4 overlies DB 5, which overlies DB 6. Data from DB 7 have been omitted from the study (see "Paleomagnetic Properties"). We assume that the Cape Lookout sites (in nearly horizontal flows exposed scores of meters above sealevel) are stratigraphically higher than the sealevel exposures of horizontal flows at Oceanside. At a small abandoned quarry 1 km south of Depoe Bay, we sampled a 15- to 20-m thick, rudely-jointed subaerial flow (sill?). The three sampling localities span approx 80 km of exposures in the Depoe Bay Basalt (fig. 7).

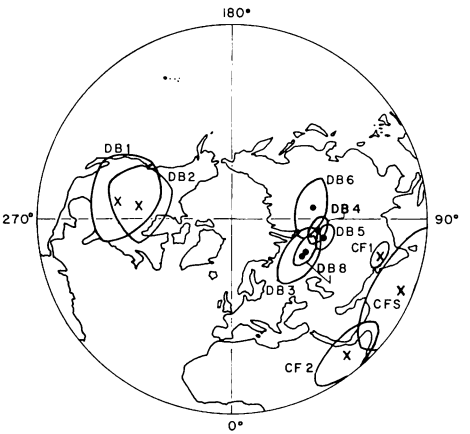


Fig. 8. Virtual geomagnetic poles (VGP) and associated confidence ovals (α_{95}) of the Cape Foulweather (CF) and Depoe Bay (DB) Basalts of the Oregon coast, as compiled in table 3; CFS = R Simpson's (written commun., 1976) Cape Foulweather pole. • = normal polarity; X = reversed polarity, transposed to the northern hemisphere.

Two groups of distinctive pole positions characterize the unit (table 3 and fig. 8). Sites at Cape Lookout (DB 3, 4, 5, and 6) and 65 km to the south at Depoe Bay (DB 8) feature nearly identical VGPs. This suggests that localized flows erupted within a short time span throughout the vent system.

The two sites at Oceanside (DB 1 and DB 2) are reversely magnetized with a pole located in the central United States (table 3 and fig. 8).

C. Significance of Results to Problems in Regional Structure

Carey (1958) has suggested that the broad arcuate structures of the Pacific Northwest are secondary features superimposed on an originally linear orogenic belt. Alternatively, Beck (1976), citing paleomagnetic poles from rocks of the western United States, maintains that the consistent trend of clockwise deflection of declinations and/or flattened inclinations has resulted from rotation and translation of microplates within a broad zone of right-lateral shear. Several workers, notably Cox (1957), Watkins (1965), Watkins and Baksi (1974), and Simpson and Cox (1977), have presented paleomagnetic evidence consistent with clockwise rotation of parts of Oregon.

The mean paleopole from 48 sites in the Yakima Basalt Subgroup (lat = 85.7°, long = 144.6°, α_{95} = 6.9°) is coincident with Beck's (1976) mean pole from undisturbed upper Tertiary rocks (lat = 85.0°, long = 113.8°, α_{95} = 5.7°), and therefore precludes significant tectonic rotation (that is, > ~5°) of southeast Washington since the middle Miocene.

Because our data from the Cape Foulweather and Depoe Bay Basalts do not approximate a Miocene pole (rotated or unrotated), we avoid speculation on the tectonic history of the northern Oregon coastal area.

D. Magnetocorrelation of Coastal and Plateau Basalts

Rietman's (ms) paleomagnetic data from 17 widely separated sites in the Frenchman Springs Member of the Wanapum Basalt in south-central Washington show 14 of those sites to be of normal polarity, while anomalous directions characterize the remainder. The similar pole positions of two anomalous Frenchman Springs flows and three sites in the Cape Foul-Weather Basalt of coastal Oregon (fig. 9), together with their similar chemistry and petrography, suggest that all these flows may have been extruded during the same geomagnetic event. The longitudinal spread of VGPs may reflect the instability of the geomagnetic field during its brief (possibly 10^2 to 10^3 yrs duration) excursion.

Depoe Bay time correlatives may also occur on the Columbia Plateau. The magnetic signature of the Depoe Bay Basalt is distinctive; the lower part of the unit (DB 1 and DB 2) has steep, negative inclinations with a corresponding VGP in the central United States, and the upper part (DB 3, 4, 5, 6, and 8) is characterized by shallow inclinations and slight westerly declinations (table 3 and fig. 8). A similar record is preserved in successive Grande Ronde flows (PG 9 and PG 8) in the Patrick Grade section, approx 500 km from the Oregon coast. We therefore tentatively correlate the Depoe Bay Basalt with these two

flows of Grande Ronde Basalt, bearing in mind that only one site (DB 6) in the younger group of Depoe Bay flows is the apparent time equivalent of PG 8 (fig. 9). We feel this correlation to be speculative, as the Depoe Bay Basalt may have been extruded during any of three transitional periods recorded by the Grande Ronde Basalt (Swanson and Wright, 1976). It seems unlikely, however, that flows extruded during different transitions would have recorded similar anomalous field behavior, since the geomagnetic pole does not necessarily follow preferred longitudinal paths during transitions (Cox, Hillhouse, and Fuller, 1975).

The paleomagnetic data for both sequences of coastal basalt, if representative, suggest that they are strictly contemporaneous with only a few plateau flows. This is an important consideration in any petrogenetic scheme attempting to account for both plateau- and locally-derived flows. Our results imply that, while plateau volcanism took place intermittently over a comparatively long period of time, only occasionally did a batch of magma make its way to the surface in coastal Oregon.

Snavely, MacLeod, and Wagner's (1973) geochemical work suggested the broadly coeval nature of coastal and Columbia Plateau volcanism. On this basis, extrusion of the Depoe Bay Basalt was confined to an interval of perhaps 2 to 3 m.y. Because of the brevity of the excursion recorded in PG 8 and 9, the paleomagnetic work may have considerably narrowed this interval.

CONCLUSIONS

The utility of the paleomagnetic method as a mapping tool, when used in conjunction with conventional stratigraphic techniques, has been

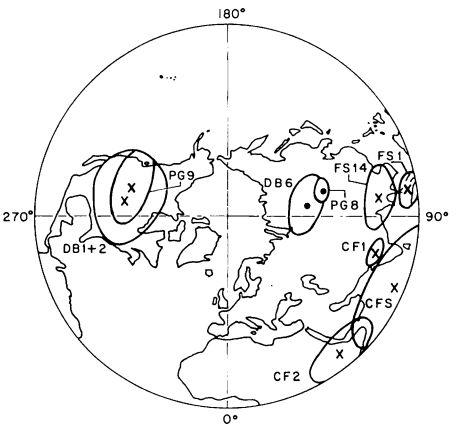


Fig. 9. Magnetocorrelation of coastal and Columbia River basalts; suggested correlation relates the Cape Foulweather (CF) Basalt of coastal Oregon (fig. 8) with two flows (FS 1 and FS 14; Rietman, ms, table 5) in the Frenchman Springs Member in south-central Washington. Second proposed correlation involves sites DB 1 and DB 2 and DB 6 in the Depoe Bay Basalt of coastal Oregon with successive flows of Grande Ronde Basalt (PG 8 and PG 9, respectively; table 2) in southeast Washington. DB 1 + 2 is mean VGP of the two sites (table 3). • = normal polarity; X = reversed polarity, transposed to the northern hemisphere.

well demonstrated by the results of this survey. The major conclusions of the study are the following:

1. Paleomagnetic data commonly permit distinction of otherwise similar flow units.

2. Individual members of the Saddle Mountains Basalt have scattered VGPs. This is expected, because the presence of erosional unconformities throughout the formation are testimony to the intermittent nature of volcanism at that time. These distinctive VGPs should prove useful in correlating isolated outcrops of many Saddle Mountains members.

3. Paleomagnetic data suggest that the Lower Monumental Member of the Saddle Mountains Basalt consists of two cooling units, although only one cooling unit occurs at any one outcrop. This suggestion needs further testing.

4. Geochemical and stratigraphic evidence had previously suggested the general equivalence of flows at all four sites in the Elephant Mountain Member of the Saddle Mountains Basalt. We have shown that flows sampled in the ancestral Snake River canyon are not precise time correlatives of those sampled on the plateau proper.

5. Our data support the hypothesis that the Pomona Member of the Saddle Mountains Basalt is a single cooling unit. The small scatter of VGPs may be a record of secular variation.

6. Paleomagnetic data and field observations suggest that the final eruptions in the Roza vent system were confined to its southern portion.

7. A sequence of Grande Ronde flows at Patrick Grade in the Blue Mountains of southeast Washington includes three successive magnetozones: N1, R2, and N2.

8. The Cape Foulweather Basalt along the Oregon coast was erupted from local volcanic centers during a geomagnetic excursion. It is tentatively considered as the time correlative of a few flows in the otherwise normally-polarized Frenchman Springs Member in south-central Washington, 400 km inland.

9. Two unusual paleopole positions characterize the Depoe Bay Basalt of coastal Oregon and permit its tentative correlation with two successive Grande Ronde flows in southeast Washington, 500 km from the coastal outcrops.

10. Southeast Washington has not undergone detectable rotational deformation (>15°) since about 15 m.y.b.p.

ACKNOWLEDGMENTS

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Appendix

Site locations

A. Columbia River Basalt Group

i. Saddle Mountains Basalt

Site	Lat. (N)	Long. (W)	Comment [Attitude]
LM1	46°32.9′	118°32.45′	roadcut
LM2	46°36.55′	117°47.0′	natural exposure
GI1	46°14.45′	118°54.9′	abandoned quarry or small roadcut
GI2	46°25.7′	119°01.6′	dike (roadcut)
MA1	46°14.25′	118°52.7′	railroad cut
MA2	46°14.5′	118°54.7′	small quarry; columnar flow
BC1	46°25.35′	119°02.0′	small mesa
BC2	46°30.75′	119°04.7′	natural exposure
BC3	46°15.55′	118°51.8′	dike cutting EM4
EM1	46°22.3′	117°03.6′	natural exposure
EM2	46°35.75′	118°32.55′	natural exposure
EM3	46°35.7′	119°00.3′	roadcut
EM4	46°15.55′	118°51.8′	railroad cut
PO1	46°35.6′	119°00.3′	roadcut
PO2	46°35.6′	118°32.85′	railroad cut
PO3	46°38.7′	117°46.1′	roadcut
PO4	46°24.8′	117°10.6′	new quarry [N80° E 5°S]
EQ1	46°35.1′	119°00.2′	roadcut
EQ2	46°35.65′	118°32.9′	natural exposure
OI1	46°35.6′	118°32.85′	railroad cut

ii. Wanapum Basalt

Site	Lat. (N)	Long. (W)	Comment
RZ1	46°19.85′	117°10.6′	roadcut
RZ2	46°15.0′	117°18.4′	natural exposure
DJ1	46°31.5′	117°48.5′	roadcut
DJ2	46°15.0′	117°45.5′	roadcut
RM1	46°17.85′	117°55.5′	town dump

iii. Grande Ronde Basalt

Site	Lat. (N)	Long. (W)	Comment
PG2	46°15.05′	117°45.3′	roadcut
PG3	46°15.0′	117°45.25′	roadcut
PG4	46°15.0′	117°45.2′	roadcut
PG5	46°14.95′	117°45.15′	roadcut
PG6	46°14.95′	117°45.2′	roadcut
PG7	46°14.98′	117°45.3′	roadcut
PG8	46°14.95′	117°45.3′	roadcut
PG9	46°14.9′	117°45.15′	roadcut
PG10	46°14.85′	117°44.95′	roadcut
PG11	46°14.9′	117°44.9′	roadcut
PG12	46°14.85′	117°44.9′	roadcut
PG13	46°14.8′	117°44.75′	roadcut
PG14	46°14.75′	117°44.7′	roadcut
PG15	46°14.75′	117°44.65′	roadcut
PG16	46°14.75′	117°44.55′	roadcut
PG17	46°14.6′	117°44.45′	roadcut
PG18	46°14.4′	117°44.5′	roadcut
PG19	46°14.3′	117°44.5′	roadcut
PG20	46°14.25′	117°44.45′	roadcut
PG21	46°14.2′	117°44.45′	roadcut
PG22	46°14.2′	117°44.5′	roadcut
PG23	46°14.2′	117°44.6′	roadcut
PG24	46°14.1′	117°44.6′	roadcut

B. Coastal basalts

Site	Lat. (N)	Long. (W)	Comment [Attitude]
CF1	44°45.35′	124°03.5′	quarry [N28°W 5°W]
CF2	44°47.1′	124°04.4′	coastal headland [N11°E 35°W]
DB1	45°27.8′	123°58.3′	coastal headland [N1°E 4°W]
DB2	45°27.8′	123°58.3′	coastal headland [N1°E 4°W]
DB3	45°21.4′	123°58.0′	roadcut [N25°E 9°W]
DB4	45°21.0′	123°58.4′	roadcut [N25°E 9°W]
DB5	45°20.9′	123°58.3′	roadcut [N25°E 9°W]
DB6	45°20.8′	123°57.9′	roadcut [N25°E 9°W]
DB8	44°47.9′	124°03.7′	quarry [NS 16°W]

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